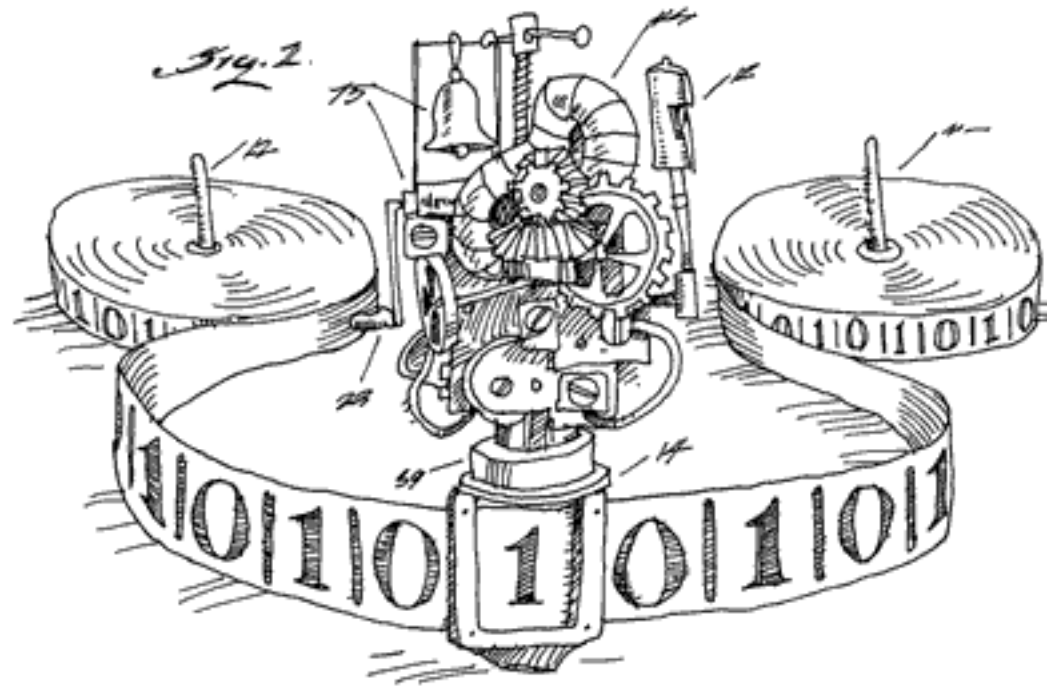


CSC4010: Introduction to Theoretical Computer Science



Lecture 8

Adapted from Sipser's MIT OCW Course on TOC

Last Time

- Is there an explicit computational problem which cannot be solved in any computational problem?
- We showed that L_{self} , and A_{TM} are undecidable via diagonalization.
- A language L is decidable if and only if L and $\bar{L}$ are both recognizable. We used this to conclude that $\bar{A}_{TM}$ is unrecognizable.
- We showed HALT_{TM} is undecidable by reducing A_{TM} to HALT_{TM} .
- **Proof Strategy:** If there is a decider H for HALT_{TM} , then we can construct a decider M' for A_{TM} . But A_{TM} is undecidable.

Reducibility Method: Undecidability of HALT_{TM}

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$M' =$ “On input $\langle M, w \rangle$

1. Simulate H on input $\langle M, w \rangle$.
2. Reject if H rejects.
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Claim: If H exists, then M' decides A_{TM} .

Reducibility - Concept

- If we have two languages (or problems) A and B , then A being reducible to B implies we can use B to solve A .
- **Example:** Measuring the area of a rectangle is reducible to measuring the lengths of its sides.
- If A is reducible to B , then solving B gives a solution to A .
 - Then, B is easy implies A is easy.
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This is the form we will use.

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Attempt 1: Given input $\langle M, w \rangle$ for M' , we pass $\langle M \rangle$ through R .

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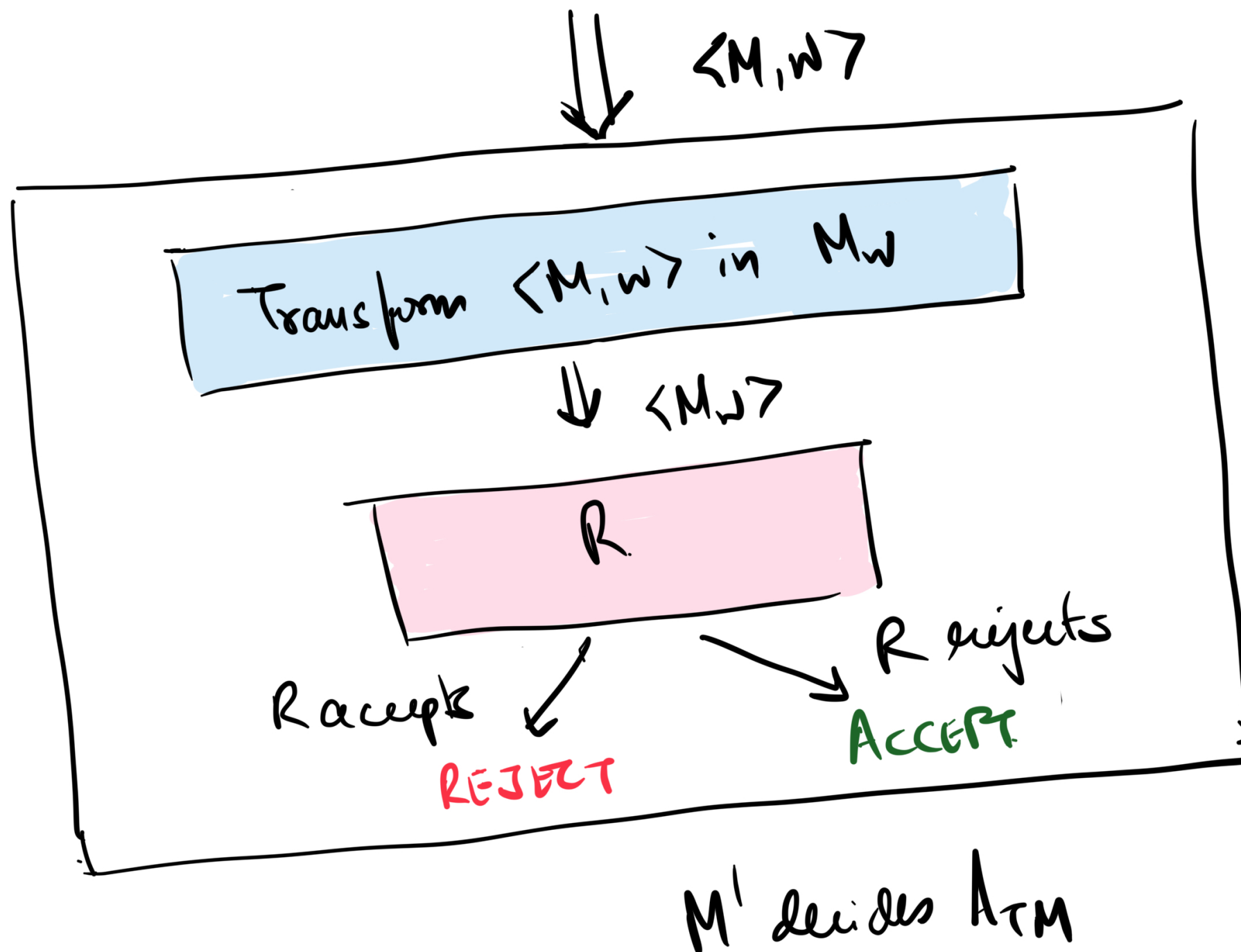
1. Transform $\langle M, w \rangle$ into M_w .
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3. **Accept** if R **rejects**.
4. **Reject** if R **accepts**.

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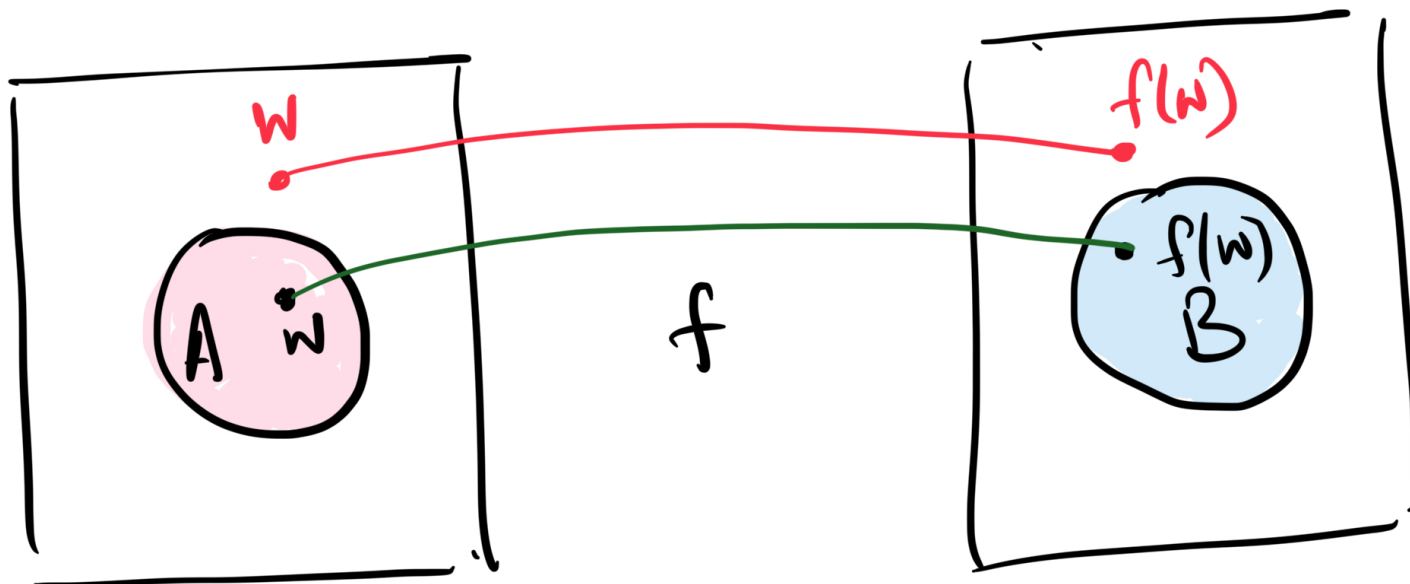
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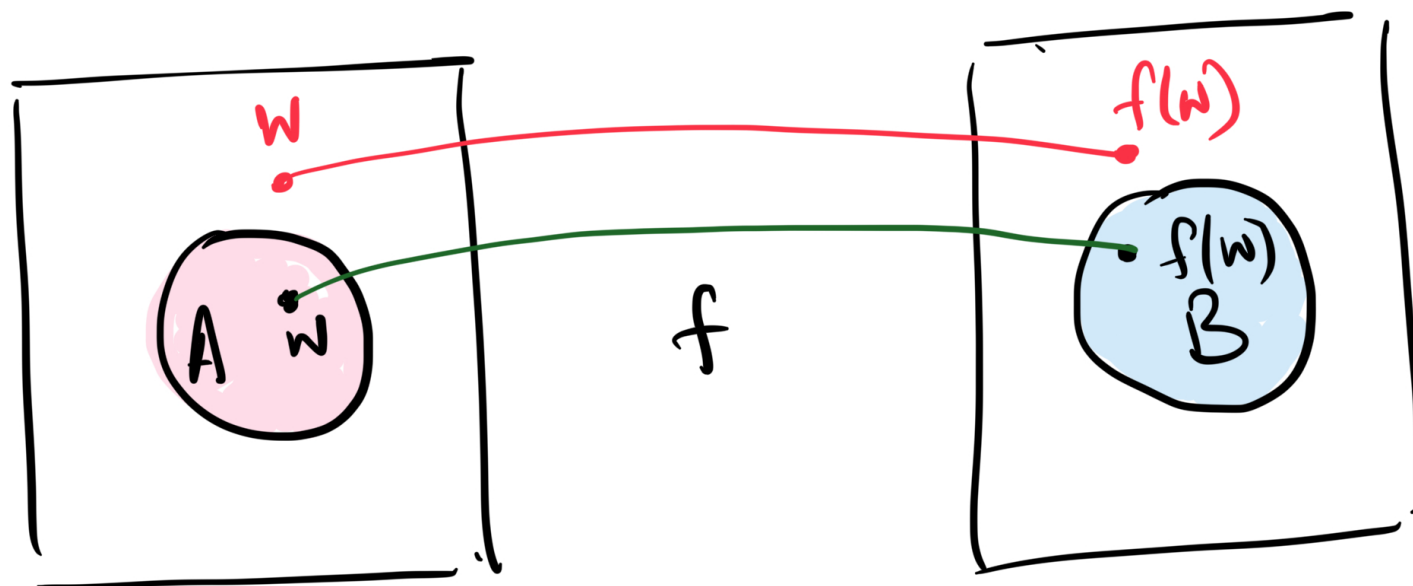
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f is called a mapping reduction from A to B .

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Claim: $f(\langle M, w \rangle) \in \overline{\text{EMPTY}}_{TM} \iff \langle M, w \rangle \in A_{TM}.$

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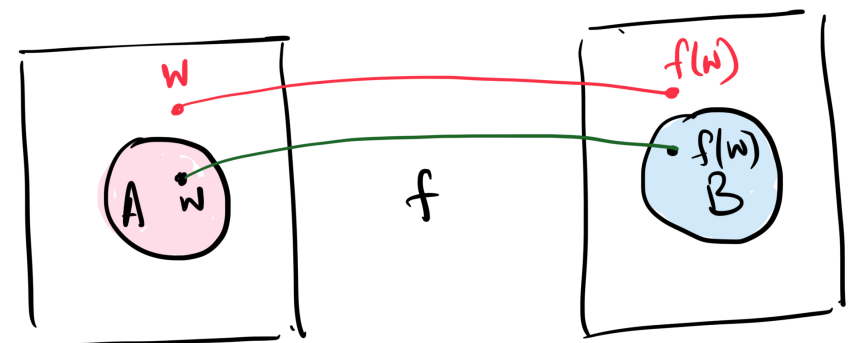
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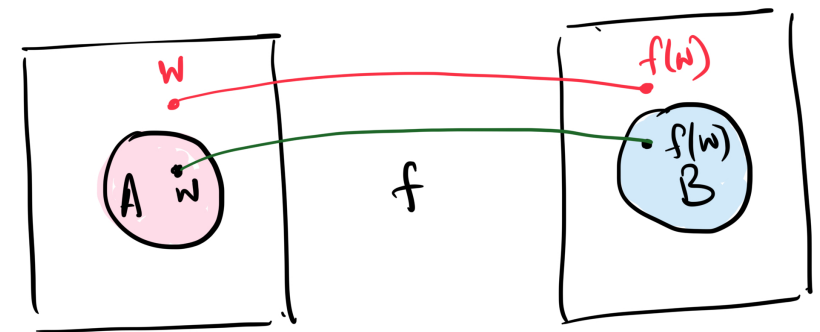
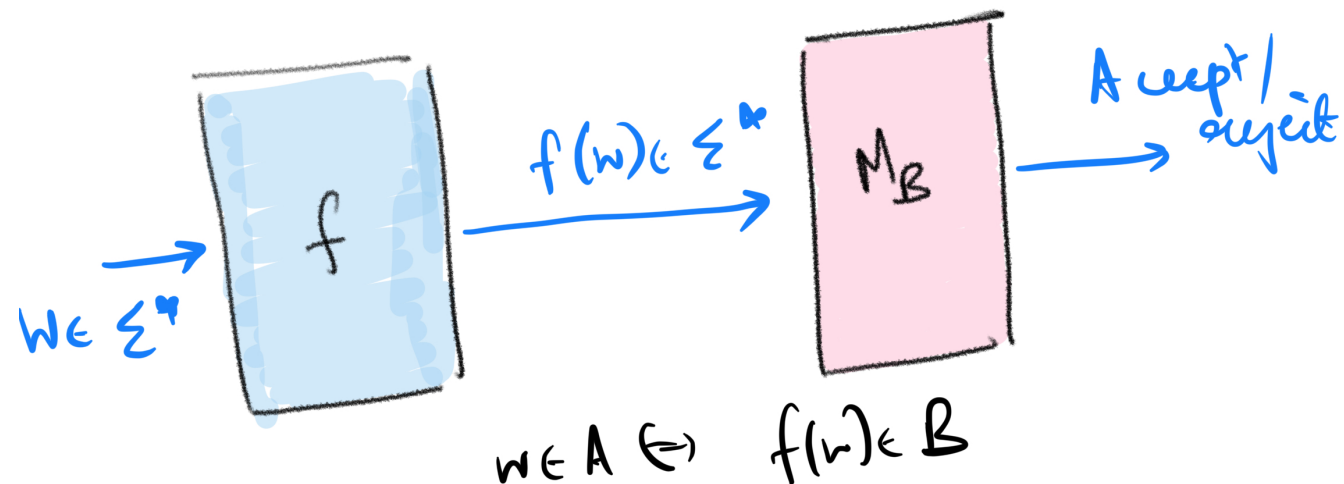
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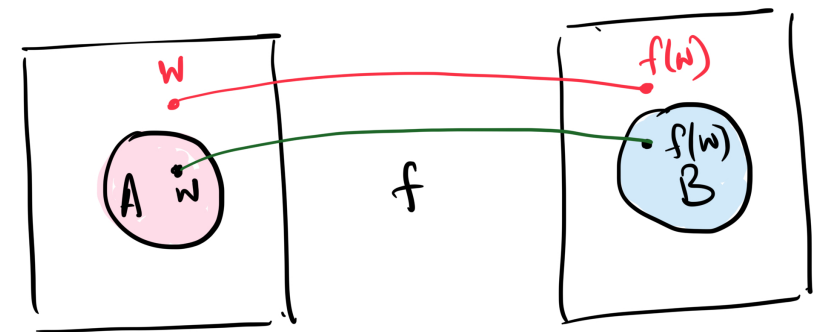
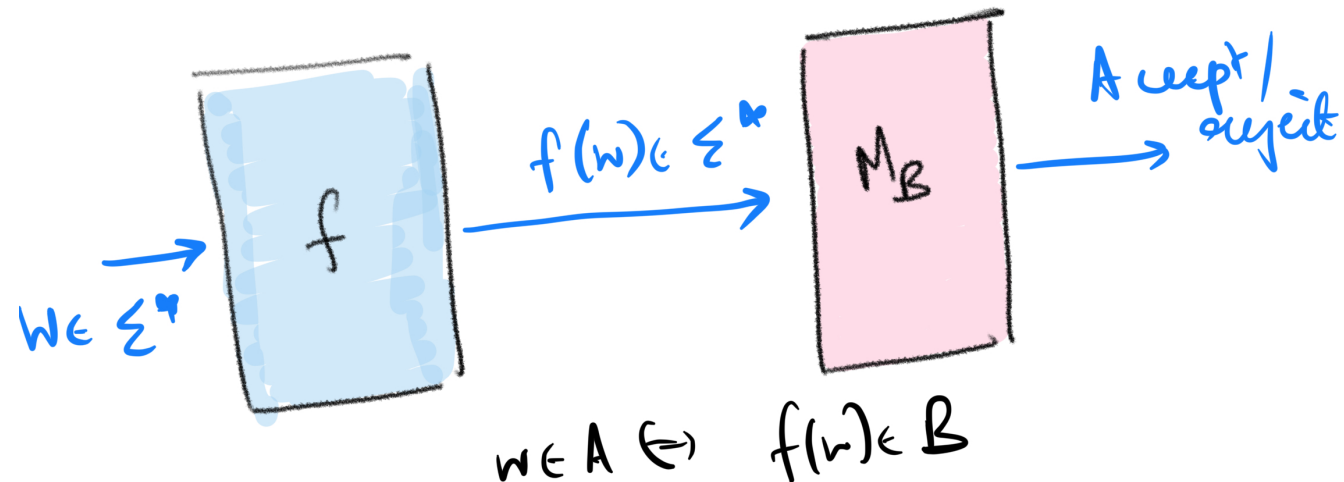
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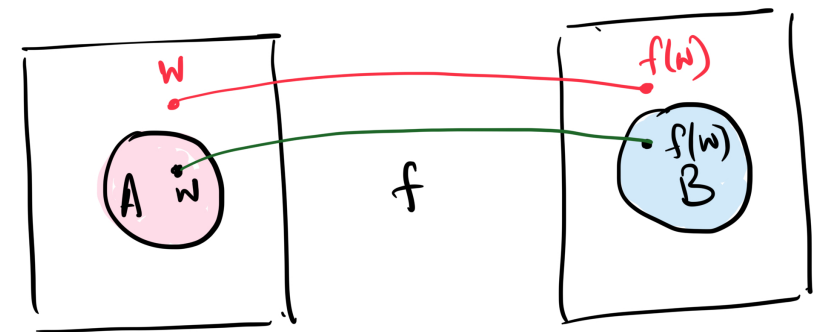
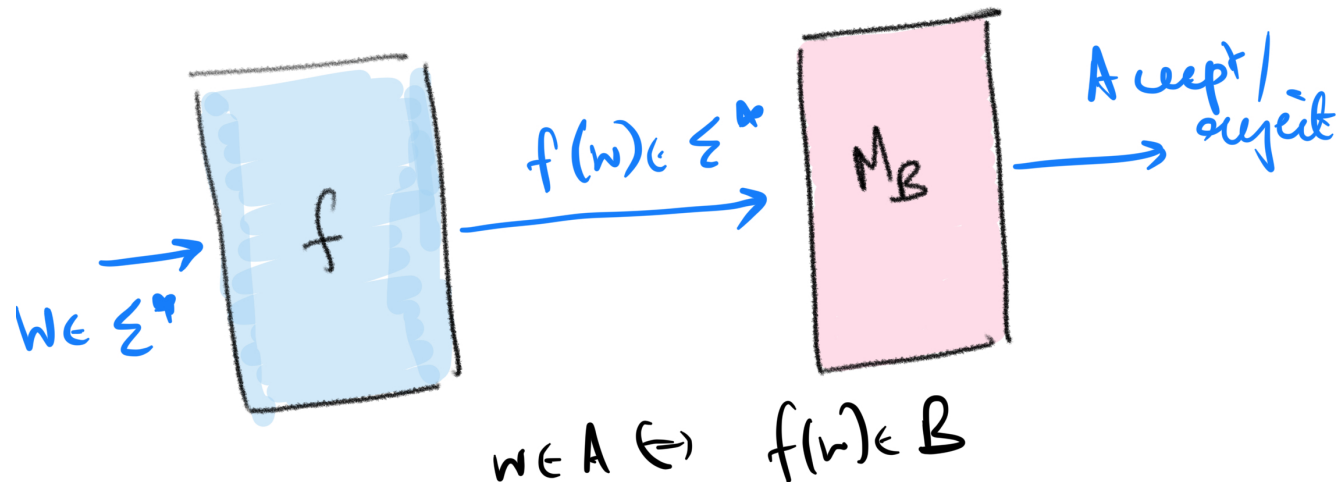


Corollary: If $A \leq_m B$ and A is undecidable, then B is undecidable.

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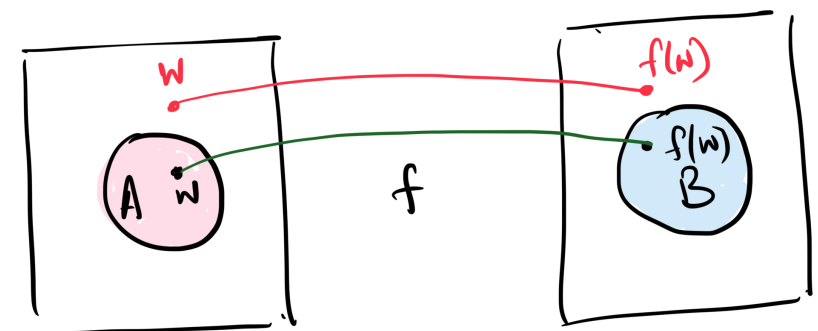
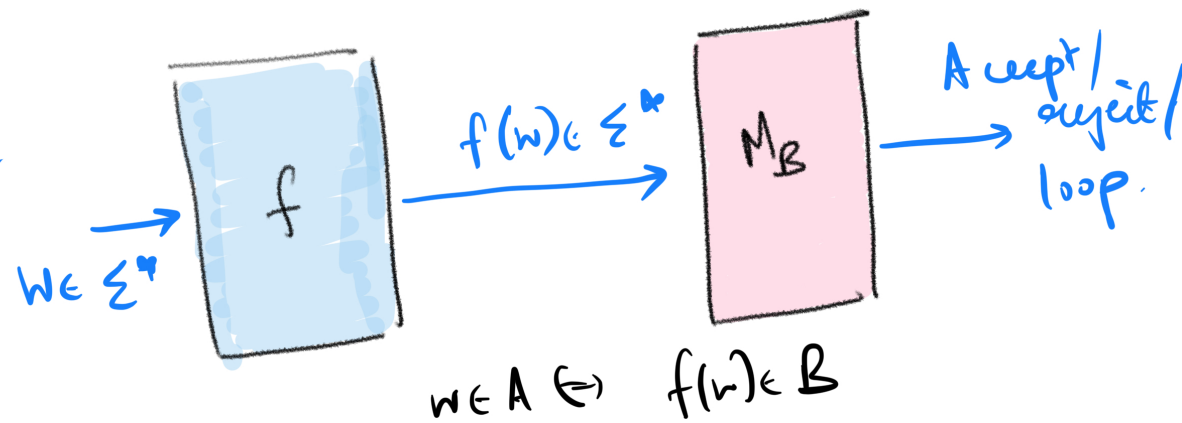
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A recipe for undecidability: To show B is undecidable, find undecidable A and a mapping reduction from A to B .

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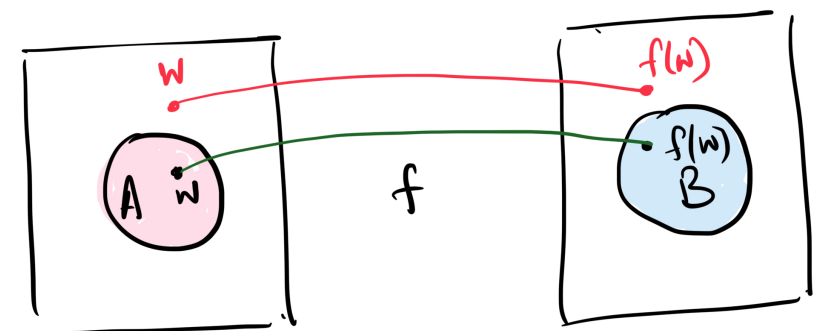
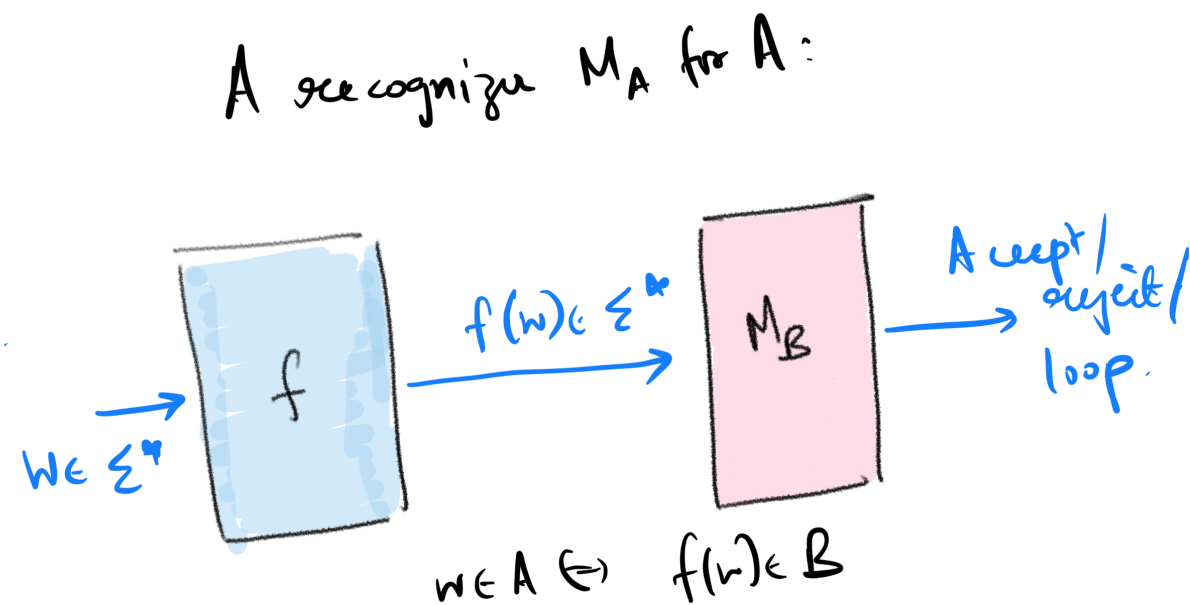
Theorem: If $A \leq_m B$ and B is recognizable, then A is recognizable.

A recognizes M_A for A :



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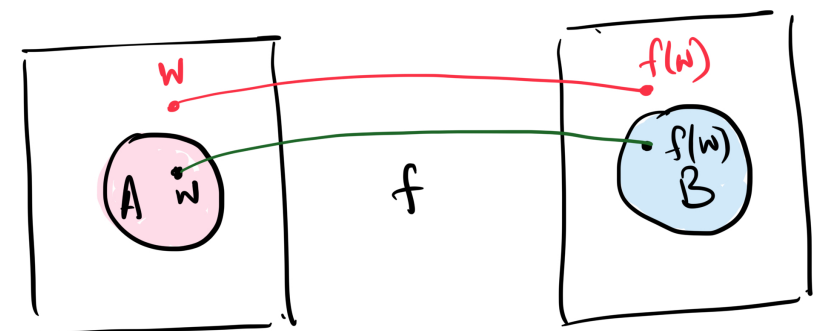
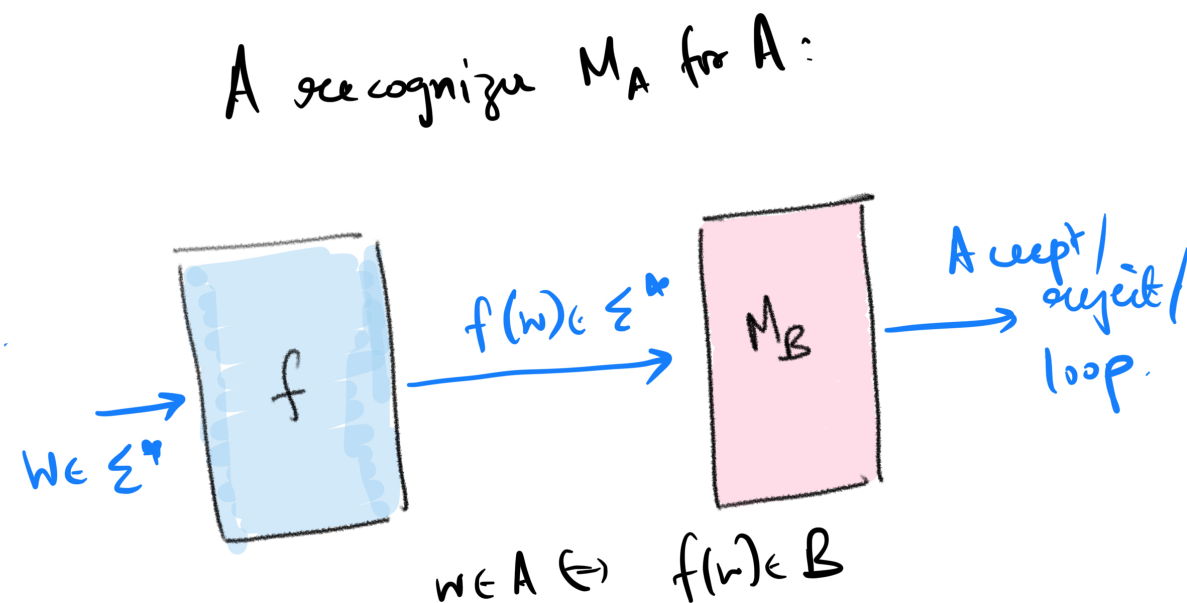
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A recipe for unrecognizability : To show B is unrecognizable, find unrecognizable A and a mapping reduction from A to B .

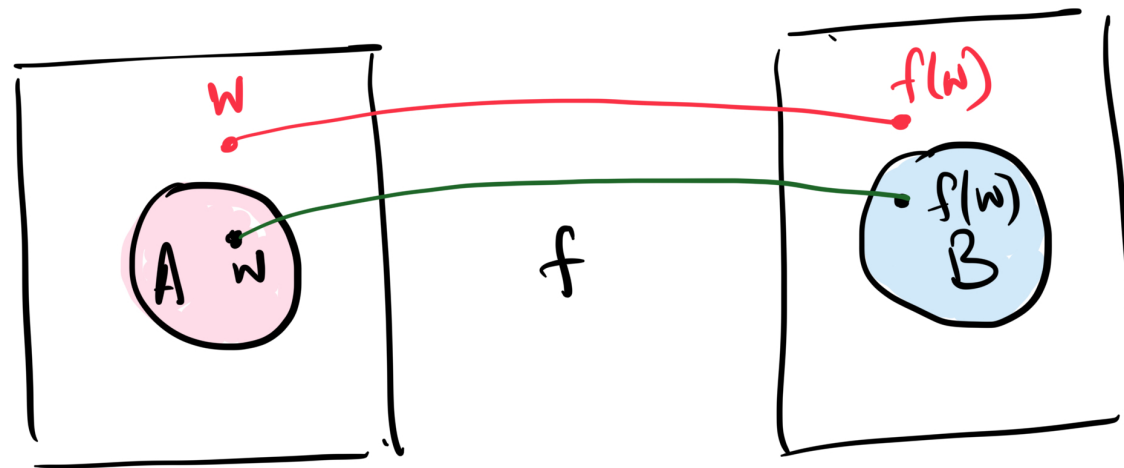
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Using Mapping Reduction for Unrecognizability

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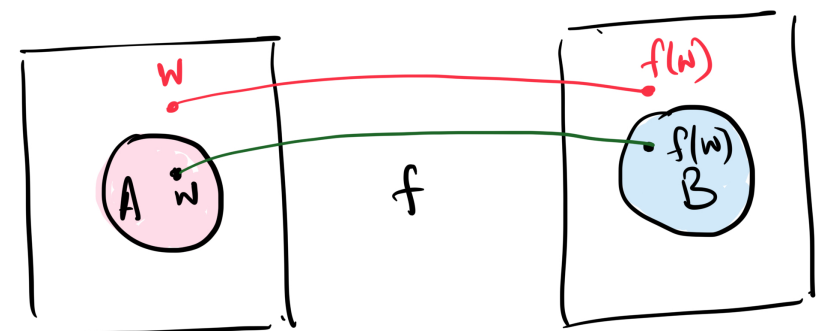
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Reducibility Terminology

Why do we use the term “reduce”?

- When we reduce A to B , we show how to solve A by using B . We conclude that A is no harder than B . (This suggests the $\leq_m$ notation).
- Possibility 1: We bring A 's difficulty down to B 's difficulty.
- Possibility 2: We bring B 's difficulty up to A 's difficulty.

Summary

- Introduced the mapping reducibility to show undecidability/unrecognizability.
- Showed EMPTY_{TM} is undecidable and unrecognizable.
- Next time: more examples.